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Geology 1988;16;654-657
doi: 10.1130/0091-7613(1988)016<0654:PRFPWC>2.3.CO;2

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Paleomagnetic results from Palau, West Caroline Islands: A constraint on Philippine Sea plate motion

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ABSTRACT

Paleomagnetic results from the Palau Islands indicate that the islands have undergone 60°–70° of clockwise rotation since mid-Oligocene time. This post–mid-Oligocene rotation is interpreted to represent the motion of the Philippine Sea plate and not a local rotation for the following reasons: (1) lack of morphologic expression of the differential rotation along the Palau-Kyushu Ridge, (2) similarity to predicted rotations obtained by using phase-shifted magnetic anomaly data, and (3) ubiquitous clockwise rotations seen in paleomagnetic data from around the Philippine Sea plate. This strong clockwise rotation of the Philippine Sea plate provides a mechanism for oblique subduction and related transcurrent motion along the margin of the Philippine archipelago.

INTRODUCTION

The Philippine Sea plate is one of the least understood plates on Earth, despite several decades of intense ship-borne geologic and geophysical studies. The plate consists of two distinct tectonic provinces, the west Philippine Sea province (WPSP) and the east Philippine Sea province (EPSP). Several models have been proposed for both provinces. Uyeda and Ben-Avraham (1972), Hilde et al. (1977), and Hilde and Lee (1984) suggested that the WPSP is a trapped piece of older Pacific plate oceanic crust, isolated from the Pacific/Kula spreading system when the Pacific plate motion changed around 44 Ma. The "trapping" was facilitated by the initiation of west-dipping subduction (proto-Mariana) along a former north-south transform fault. In contrast, Lewis et al. (1982) suggested that the WPSP originated by back-arc spreading behind a northeast-dipping subduction zone.

Karig (1971, 1975) first proposed that the EPSP is a series of remnant arcs and back-arc basins, formed behind the Mariana subduction system. Since then, many variations have been proposed (Scott et al., 1981; Uyeda and McCabe, 1983), but the concept of "arc-splitting" and back-arc spreading is central to most models. The origin and history of the Phil-

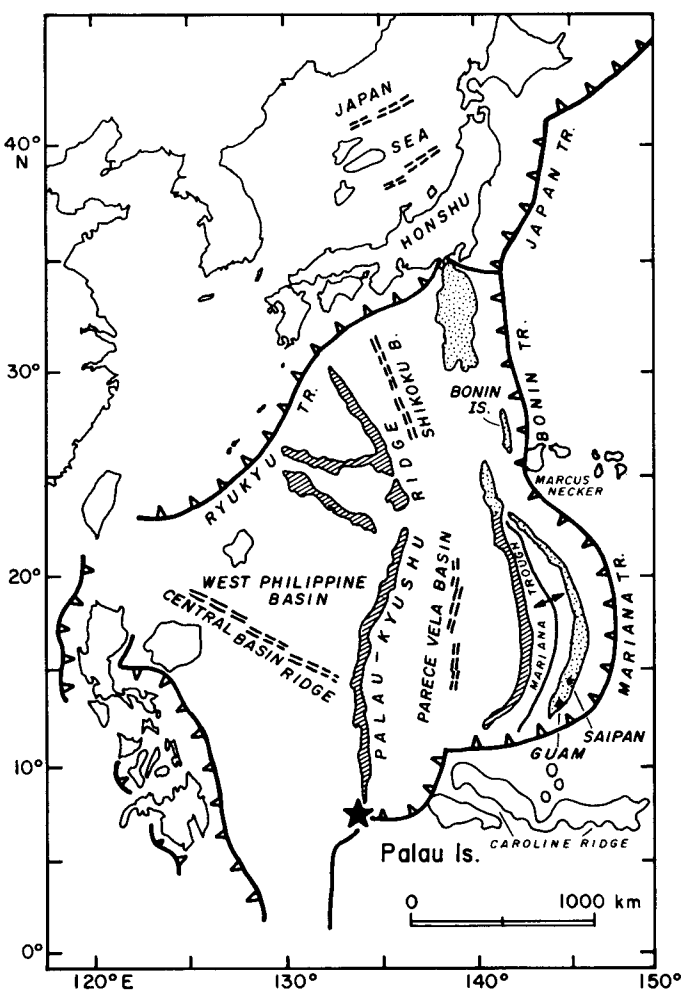


Figure 1. Major features of Philippine Sea plate.

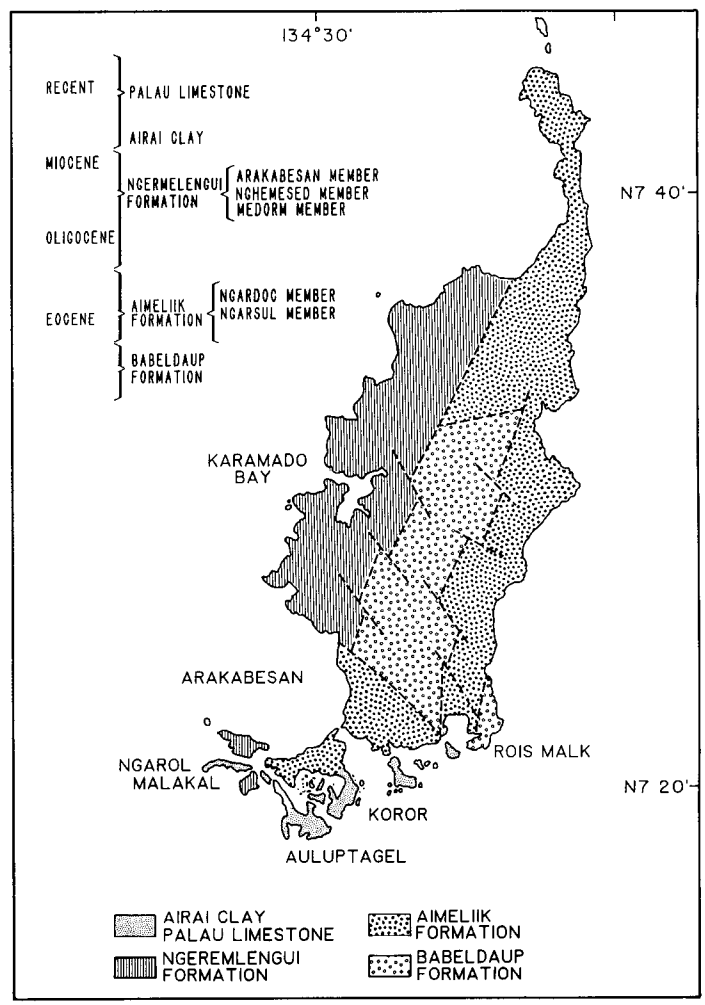


Figure 2. Generalized geologic map of northern Palau Islands, adapted from Mason et al. (1956).

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ippine Sea plate has emerged as an important and unresolved aspect of the tectonics of the Southeast Asian accretionary margin.

The Palau Islands are located on the southern end of the Palau-Kyushu Ridge, which marks the eastern extent of the west Philippine Sea basin (Fig. 1). Because the Palau Islands have been associated with the WPSP since Eocene time, they offer a unique paleomagnetic opportunity to determine the past motion of the Philippine Sea plate.

The history of Philippine Sea plate motion is poorly known. Determination of its present motion is difficult because the plate is surrounded by subduction zones. Nevertheless, relative to Eurasia, the plate appears to have an overall clockwise rotation at a rate of 1.5° to 2.0°/m.y. around a pole of rotation just north of Tokyo (Ranken et al., 1984). How far this motion can be extrapolated into the past is unknown. Other constraints on past motion are (1) sea-floor magnetic lineation (Lee and Hilde, 1971; Louden, 1976; Shih, 1980; Hilde and Lee, 1984); (2) inclination data from Deep Sea Drilling Project (DSDP) drill cores (Louden, 1977; Bleil, 1981); (3) paleomagnetic data from the Mariana arc (Fuller et al., 1980; M. Fuller, unpub. data; Keating and Helsley, 1985; Kodama et al., 1983; and others); and (4) preliminary paleomagnetic results from the Palau Islands (Aoki, 1972). The interpretation of these data remains controversial. We have therefore undertaken a paleomagnetic survey of the Palau Islands to get an independent test for the past motion of the Philippine Sea plate and models of its origin.

GEOLOGY OF PALAU

The Palau Islands are composed of island-arc subaerial and submarine volcanic rocks and reef limestones ranging in age from Eocene to Holocene. The northern islands are predominantly volcanic, whereas those in the south contain Pleistocene to Holocene reef limestones. Mason et al. (1956) established the relative stratigraphic relation between the rock units, and they divided the volcanic section into three units: in order of decreasing age, these are the Babeldaup Formation, the Aimeliik Formation (encompassing the Ngardok, Ngarsul, and Galap Tuff Members), and the Ngermelengui Formation (containing the Nghemesed, Medorm, and Arakabesan Members). Overlying the volcanic section is the Miocene to Pliocene Airai Clay and the Miocene to Pleistocene Palau Limestone (Fig. 2).

Some preliminary K/Ar age determinations were made on the volcanic section by Meijer et al. (1983). Two radiometric ages on clasts from the Ngeremlengui Formation indicated an early Miocene (20.1 ±0.5 Ma) age for the youngest Arakabesan Member and a mid-Oligocene age (30.3 ± 0.9 Ma) for the older Medorm Member. One radiometric date (32.3 ±1.1 Ma) has been obtained from a hornblende concentrate of an altered lapilli tuff from the Aimeliik Formation, but this early Oligocene date does not agree with late Eocene fossils from the same location (Cole, 1950). However, a whole-rock K/Ar date (this study) obtained from a fresh andesite flow in the Aimeliik Formation yields an age of 32.4 ±1.8 Ma. The concordance of the whole-rock and hornblende concentrate ages suggests that the Aimeliik Formation is indeed mid-Oligocene in age. A single whole-rock K/Ar date (this study) from the Babeldaup Formation indicates a late Eocene age for this formation (37.7 ±3.1 Ma).

TABLE 1. SITE RESULTS FROM THE PALAU ISLANDS

Site no.	No. of Samples	Demag Level*	In-situ Inc (°)	In-situ Dec (°)	Structural Correction (°)	Corrected Inc (°)	Corrected Dec (°)	K	α_{95}
Arakabesan Member: early Miocene (radiometric age 20.1 Ma)									
PA86-4	7	A/200-600†	-21.6	49.8	170/32 W	5.7	50.8	44.8	9.1
PA86-5	7	A/ SC	-18.5	52.3	170/32 W	16.2	52.8	391.3	3.0
PA86-16	7	A/ SC	18.9	246.0	014/41 W	-2.1	243.4	403.6	3.1
PA86-17	10	A/T SC	39.3	257.2	014/41 W	18.1	249.1	57.0	6.5
PA86-24	6	A/ SC	11.2	216.5	098/25 S	-10.8	216.5	404.7	3.3
Medorm Member: middle Oligocene (radiometric age 30.3 Ma)									
PA86-10	6	A/ SC	-26.0	250.4	110/30 S	-8.3	242.6	784.9	2.4
PA86-23	7	A/ SC	8.9	67.7	055/32 S	2.1	70.7	181.6	4.5
Aimeliik Formation: early Oligocene (radiometric age 32.3 Ma)									
PA86-20	13	A/T400-900	-46.1	94.1	048/36 N	-20.9	109.6	8.1	15.4
PA86-22	5	A/400-500	42.8	257.3	218/45 W	9.6	273.8	44.5	11.6
PA86-21&22§	6	A/400-600	-0.2	282.4	218/45 W	-34.8	275.0	48.0	9.8
Babeldaup Formation: late Eocene (radiometric age 37.7 Ma)									
PA86-13	7	A/T SC	10.2	081.9	218/22 W	24.2	76.6	38.4	9.9
PA86-19	6	A/T SC	-0.3	238.8	105/25 S	-18.2	241.4	67.3	8.2

* A = AF demagnetization; SC = single component of magnetization; T = stepwise thermal demagnetization
† Demagnetization values used in selection of stable direction.
§ Remagnetized samples.

PALEOMAGNETIC RESULTS

A total of 278 samples from 24 sites were collected from Babeldaup, Koror, Arakabesan, and Malakal islands. Because of the poor exposure of other units, samples were predominantly from the Aimeliik and Ngermelengui Formations (Fig. 2). Samples from all these sites were analyzed by using alternating field (AF) and stepwise thermal demagnetization techniques.

Each sample was progressively demagnetized and then measured with a Molspin fluxgate-type spinner magnetometer. Stable remanent magnetization directions were selected by using equal-area and Zijderveld (1967) orthogonal demagnetization plots. Site averages were then computed by using Fisher (1953) statistics (Table 1).

Of the 24 sites measured, 12 yielded consistent stable directions. Five sites from the early Miocene Arakabesan Member yielded a mean declination deflection of 54° ±17°. Two sites from the mid-Oligocene Medorm Member had a mean declination deflection of 66°. Three sites from the early Oligocene Aimeliik Formation had a mean declination of 101° ±20°, and two sites from the late Eocene Babeldaup Formation yielded a mean declination of 69°. All of these sites showed a strong clockwise deflection of both the in situ and tilt-corrected direction (Fig. 3). Despite the relatively small number of sites, both normal and reversed directions were found in all the units sampled.

All but three sites (PA86-21 and 22, PA86-13, and PA86-19) showed excellent magnetic behavior and were fresh and unaltered. The dates of magnetization of these sites are interpreted to be equivalent to the ages of the rocks. The three sites that require further interpretation are discussed below.

Sites PA86-21 and PA86-22 are located in a section of interbedded, fine-grained, water-lain tuffs and turbidites from the Ngardok Member of the Aimeliik Formation. The results from these two sites are discussed together because they are stratigraphically only 5 m apart. Several beds were sampled within each site. Beds that appear more altered yield significantly different tilt-corrected directions from those that are unaltered. However, the in situ direction of the altered beds is concordant with the tilt-corrected directions of the unaltered units (Table 1). We interpret the magnetization of the unaltered rocks to record a primary early Oligocene direction. The altered beds are interpreted as an early Oligocene remagnetization. We assign an early Oligocene age to this remagnetization because of the concordance of the altered in situ direction and the unaltered tilt-corrected direction. Note that both the in situ and tilt-corrected directions are strongly deflected, yielding similar declination anomalies.

Sites PA86-13 and PA86-19 are from the late Eocene Babeldaup Formation. Both sites show similar in situ and tilt-corrected directions. However, there are several reasons to suspect that the date of magnetization and age of the rocks may not be equivalent: (1) The in situ

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directions are indistinguishable from the middle Oligocene results. In recent years, paleomagnetic studies from tectonically active areas have increasingly demonstrated that when coincident directions are found in units of different age, remagnetization of the older unit is likely (e.g., Moreau et al., 1987; Almasco et al., 1988). (2) The tilt-corrected directions indicate a paleolatitude of around 10°N. This is hard to reconcile with the early Oligocene results from this study as well as the DSDP inclination data (Louden, 1977; Bleil, 1981). (3) Site PA86-13 is located in a zone of mineralization and extreme alteration (Mason et al., 1956; Rytuba et al., 1986). (4) Rock magnetic experiments suggest that the rocks are remagnetized (Haston, 1988). Therefore, it is not clear whether the results from the Babeldaup represent a primary late Eocene direction or a younger remagnetization event.

DISCUSSION

The paleomagnetic data from the Palau Islands suggest the following: (1) Since the middle Oligocene, the islands have undergone a clockwise rotation of about 60°–70°. (2) The Palau Islands may have undergone a short period (2–3 m.y.) of rapid clockwise rotation between early and mid-Oligocene time. However, the rapidity of this rotation hinges upon a literal interpretation of the age of the rocks. Despite the low latitude of the islands, which could give rise to an equatorial ambiguity, the sense of rotation can be seen to be clockwise by comparing the Miocene and early Oligocene results (Fig. 3). Results from Aoki (1972) seem to agree with this study, but those results are not included because of their preliminary nature. These data can be interpreted in three basic ways: (1) localized rotation of the Palau Islands, (2) rotation of the Philippine Sea plate, and (3) a combination of local and plate-wide rotations.

Because the Palau Islands are located on a plate margin near a tectonically complex area, local rotation of the area is possible. However, there are several reasons to suspect that the mid-Oligocene and younger results do not reflect local rotation: (1) In the vicinity of the Palau Islands, the morphological trend of the Palau-Kyushu Ridge does not reflect the 70°–90° of rotation indicated by the paleomagnetic data. (2) The rocks and structures on the Palau Islands are those associated with the formation of the island arc (Mason et al., 1956). Therefore, any tectonic event that may have caused a local rotation of the Palau Islands is not reflected in the geology. (3) The paleomagnetic data are similar to predicted rotations obtained by using phase-shifted magnetic anomaly data from the west Philippine Sea basin (Louden, 1977; Shih, 1980; Hilde and Lee, 1984). (4) Paleomagnetic data from around the Philippine Sea plate show ubiquitous clockwise rotations. Data from Guam (Larson et al., 1975; M. Fuller, unpub. data), Saipan (Fuller et al., 1980), and the Bonin Islands (Kodama et al., 1983; Keating and Helsley, 1985) all show strong clockwise rotations.

Prior to this study, paleomagnetic data from the Mariana arc have been interpreted as reflecting deformation along the arc; i.e., when the Mariana arc became pinned at the ends by the Marcus-Necker and Caroline ridges, continued back-arc spreading caused deformation of the arc (Larson et al., 1975; McCabe and Uyeda, 1983; McCabe, 1984). However, because the Philippine Sea plate appears to have rotated 60°–70° since the mid-Oligocene, no special mechanism is required to explain rotations along the Mariana arc.

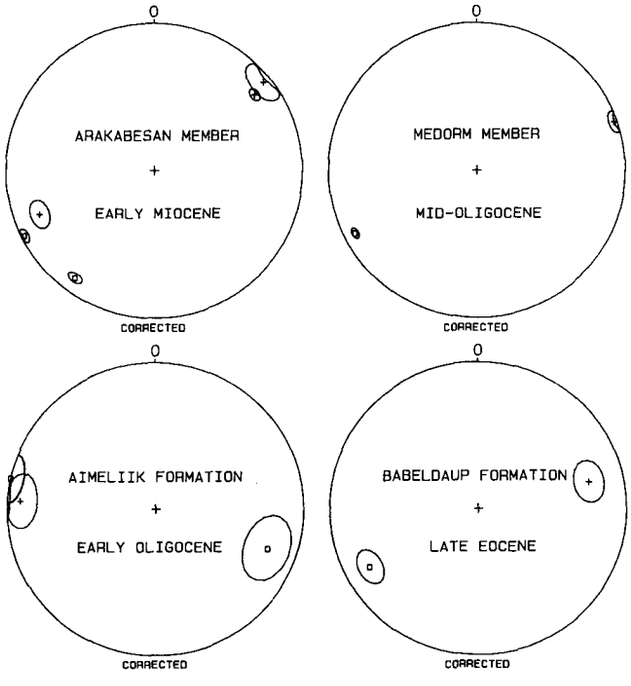


Figure 3. Site results from Palau Islands with associated 95% confidence intervals. Crosses = positive inclinations; circles = negative inclinations.

The rapid 30°–40° of clockwise rotation between early and mid-Oligocene time may be local rather than plate-wide. What mechanism might account for a short, rapid rotation? Mrozowski and Hayes (1979) showed that the Parece Vela basin opened between 30 and 17 Ma. It is possible that in the process of arc-splitting, tectonism started sometime before the oldest magnetic anomalies seen in the back-arc basin. Moreover, at the time of the oldest anomaly, the arc splitting had already been completed. With this in mind, it is plausible that the rapid rotation between 32 and 30 Ma corresponds to the onset of rifting in the Parece Vela basin. However, the data suggesting the 32 to 30 Ma rotation are sparse.

The results from the upper Eocene Babeldaup Formation are enigmatic. If they were remagnetized in the mid-Oligocene, then they fit the above scenario. However, if they record a late Eocene direction, a much more complex tectonic history is required. In this case, the Palau Islands were rotated counterclockwise 30°–40° and translated from 10°N to 2°S between 37 and 32 Ma. At 32 Ma, this scenario merges with the one above. However, this interpretation is counter to DSDP data (Louden, 1977; and others) and requires a complex rotation history for which no known mechanisms exist. We prefer the mid-Oligocene remagnetization hypothesis, for which there is some rock magnetic justification (Haston, 1988).

Is this large rotation of the Philippine Sea plate consistent with the geology and structures around the margin of the plate? Karig (1983), Sarewitz and Karig (1986), and Karig et al. (1986) proposed that several large strike-slip faults within the Philippine archipelago have transported tectonostratigraphic terranes large distances from origins south of their present positions. A clockwise rotating plate on the eastern boundary of the Philippines produces a natural mechanism for driving large lateral motion behind an oblique subduction zone. This is similar to the ideas proposed by Fitch (1972) and Beck (1980). Modern analogies are found in the Sumatra–Andaman Sea region (Fitch, 1972), as well as in the Philippines. Furthermore, the disrupted island-arc rocks found along the eastern margin of the Philippines (Moore et al., 1981a, 1981b; Hawkins et al., 1986) may be accreted bits of an island arc originally from the Philippine Sea plate.

CONCLUSIONS

The paleomagnetic results from the Palau Islands indicate a two-phase history of rotation. The first phase, which is poorly constrained, happened between 32 and 30 Ma and was a period of rapid (15°–20°/m.y.) clockwise rotation (30°–40°). This rotation is possibly associated with the onset of rifting in the Parece Vela basin. The next phase was a period of slower (2°–3°/m.y.) prolonged clockwise rotation

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60°–70°). We interpret this period of rotation to reflect the motion of the Philippine Sea plate since the mid-Oligocene. The rate of rotation is comparable to that seen at present.

The large rotation of the Philippine Sea plate since the mid-Oligocene requires modification of existing models for its origin and history. The "trapped" crust model (Uyeda and Ben-Avraham, 1972; Hilde and Lee, 1984) requires large north-south–oriented transform faults. However, when the rotation of the Philippine Sea plate is taken into account, the spreading center in the WPSP was oriented northeast-southwest, roughly parallel to the margin of Asia. Therefore, it seems more likely that the WPSP has a back-arc origin. The rotation of the Philippine Sea plate may have acted as the driving mechanism for translation of terranes in the Philippines.

Further work is continuing on Guam and Saipan to determine their rotational history and to see whether they rotated with the WPSP. Ultimately, we would like to combine the paleomagnetic, magnetic anomaly, and DSDP inclination data to try to determine a pole of rotation for the Philippine Sea plate. With the advent of oriented drill cores (i.e., Halgedahl and Jarrard, 1987), it may now be possible to get paleomagnetic data from drill cores located away from the plate margins; this will certainly eliminate problems with local rotations caused by the deformation of island arcs.

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ACKNOWLEDGMENTS

Supported by National Science Foundation Grant EAR 84-08354 and generous donations from an oil company consortium. We thank the Palau Government and the people of Palau for their gracious help and hospitality while we conducted this research; J. R. Dunn for his superb work in the laboratory; M. Reagan for useful guidance; and V. Courtillot for his input and good ideas.

Manuscript received October 12, 1987
Revised manuscript received March 16, 1988
Manuscript accepted April 21, 1988